

$$\sqrt{0.7117}$$

$$= 2.33$$

5 (a)(i)

$$F_A = q_A E$$

$$= 2(1.6 \times 10^{-19})(1.5 \times 10^5)$$

$$= 4.8 \times 10^{-14} \text{ N (to the right)}$$

(a)(ii)

$$F_{A \text{ due to B}} = q_A E_B$$

$$= 2(1.6 \times 10^{-19}) \frac{(-2 \times 1.6 \times 10^{-19})}{4\pi(8.85 \times 10^{-12})(4.0 \times 10^{-12})^2}$$

$$= -5.8 \times 10^{-5} \text{ N (at angle of } 60^\circ \text{ below horizontal towards B)}$$

(b)

Since charge B is equal in magnitude to charge A but opposite in sign, the force on charge B due to the electric field is equal in magnitude but acts horizontally in the opposite direction to the electric field. Hence resultant force on molecule AB due to the electric field is zero.

Also, the electrostatic force on charge B due to charge A ($F_{B \text{ due to A}}$) is the reaction pair to the electrostatic force on charge A due to charge B ($F_{A \text{ due to B}}$).

(c)	They both share the same magnitude but act in opposite directions. Hence the resultant electrostatic force on molecule AB is zero. As such the resultant force on molecule AB is zero. Torque = Fd $= 4.8 \times 10^{-14} \sin 60^\circ (4.0 \times 10^{-12})$ $= 1.66 \times 10^{-25} \text{ Nm (clockwise)}$
6 (a)	${}_{54}^{128}\text{D}$